

13-3c Cost Curves and Their Shapes

Just as we found graphs of supply and demand useful when analyzing the behavior of markets in previous chapters, we will find graphs of average and marginal cost useful when analyzing the behavior of firms. [Figure 4](#) graphs Caleb's costs using the data from [Table 2](#). The horizontal axis measures the quantity the firm produces, and the vertical axis measures marginal and average costs. The graph shows four curves: average total cost (ATC), average fixed cost (AFC), average variable cost (AVC), and marginal cost (MC).

Figure 4

Caleb's Average-Cost and Marginal-Cost Curves

This figure shows the average total cost (ATC), average fixed cost (AFC), average variable cost (AVC), and marginal cost (MC) for Caleb's Coffee Shop. All of these curves are obtained by graphing the data in [Table 2](#). These cost curves show three common features: (1) Marginal cost rises with the quantity of output. (2) The average-total-cost curve is U-shaped. (3) The marginal-cost curve crosses the average-total-cost curve at the minimum of average total cost.

The cost curves shown here for Caleb's Coffee Shop have some features that are common to the cost curves of many firms in the economy. Let's examine three features in particular: the shape of the marginal-cost curve, the shape of the average-total-cost curve, and the relationship between marginal cost and average total cost.

Rising Marginal Cost

Caleb's marginal cost rises as the quantity of output produced increases. This upward slope reflects the property of diminishing marginal product. When Caleb produces a small quantity of coffee, he has few workers, and much of his equipment is not used. Because he can easily put these idle resources to use, the marginal product of an extra worker is large, and the marginal cost of producing an extra cup of coffee is small. By contrast, when Caleb produces a large quantity of coffee, his shop is crowded with workers, and most of his equipment is fully utilized. Caleb can produce more coffee by adding workers, but these new workers have to work in crowded conditions and may have to wait to use the equipment. Therefore, when the quantity of coffee produced is already high, the marginal product of an extra worker is low, and the marginal cost of producing an extra cup of coffee is large.

U-Shaped Average Total Cost

Caleb's average-total-cost curve is U-shaped, as shown in [Figure 4](#). To understand why, remember that average total cost is the sum of average fixed cost and average variable cost. Average fixed cost always declines as output rises because the fixed cost is getting spread over a larger number of units. Average variable cost usually rises as output increases because of diminishing marginal product.

Average total cost reflects the shapes of both average fixed cost and average variable cost. At very low levels of output, such as 1 or 2 cups per hour, average total cost is very high. Even though average variable cost is low, average fixed cost is high because the fixed cost is spread over only a few units. As output increases, the fixed cost is spread over more units. Average fixed cost declines, rapidly at first and then more slowly. As a result, average total cost also declines until the firm's output reaches 5 cups of coffee per hour, when average total cost is \$1.30 per cup. When the firm produces more than 6 cups per hour, however, the increase in average variable cost becomes the dominant force, and average total cost starts rising. The tug of war between average fixed cost and average variable cost generates the U-shape in average total cost.

The bottom of the U-shape occurs at the quantity that minimizes average total cost. This quantity is sometimes called the [efficient scale \(the quantity of output that minimizes average total cost\)](#) of the firm. For Caleb, the efficient scale is 5 or 6 cups of coffee per hour. If he produces more or less than this amount, his average total cost rises above the minimum of \$1.30. At lower levels of output, average total cost is higher than \$1.30 because the fixed cost is spread over so few units. At higher levels of output, average total cost is higher than \$1.30 because the marginal product of inputs has diminished significantly. At the efficient scale, these two forces are balanced to yield the lowest average total cost.

The Relationship between Marginal Cost and Average Total Cost

If you look at [Figure 4](#) (or back at [Table 2](#)), you will see something that may be surprising at first. *Whenever marginal cost is less than average total cost, average total cost is falling. Whenever marginal cost is greater than average total cost, average total cost is rising.* This feature of Caleb's cost curves is not a coincidence from the particular numbers used in the example: It is true for all firms.

To see why, consider an analogy. Average total cost is like your cumulative grade point average. Marginal cost is like the grade you get in the next course you take. If your grade in your next course is less than your grade point average, your grade point average will fall. If your grade in your next course is higher than your grade point average, your grade point average will rise. The mathematics of average and marginal costs is exactly the same as the mathematics of average and marginal grades.

This relationship between average total cost and marginal cost has an important corollary: *The marginal-cost curve crosses the average-total-cost curve at its minimum.* Why? At low levels of output, marginal cost is below average total cost, so average total cost is falling. But after the two curves cross, marginal cost rises above average total cost. As a result, average total cost must start to rise at this level of output. Hence, this point of intersection is the minimum of average total cost. As we will see in the next chapter, minimum average total cost plays a key role in the analysis of competitive firms.

Chapter 13: The Costs of Production: 13-3c Cost Curves and Their Shapes

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.